

Optimizing Facial Recognition Models to Improve Power and Speed Efficiency

Alphanso Davis-Moses



- Supply Chain and Data Analytics
- Business Perspective

Simion Cartis



- Computer Science
- Developer

Mabel Hurd



- Computer Science
- Developer, Research

Our diverse backgrounds allow us to approach the problem from different angles and build a more comprehensive solution.

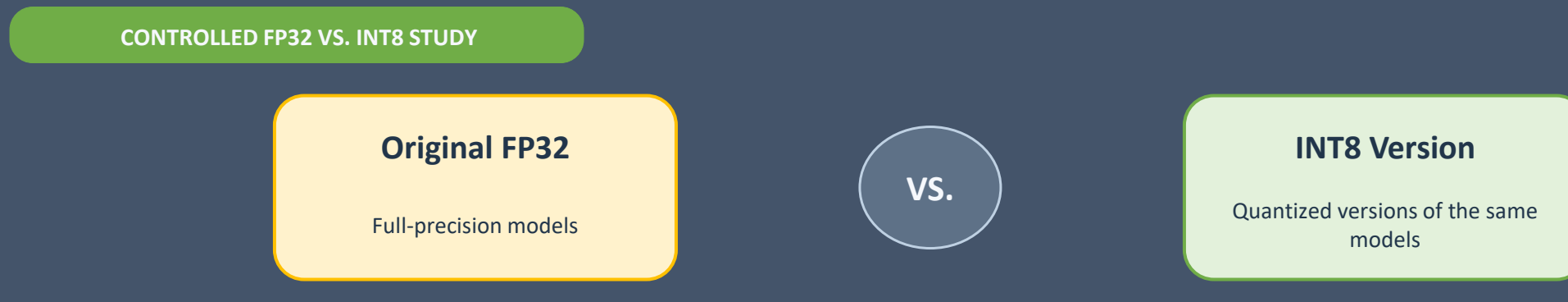
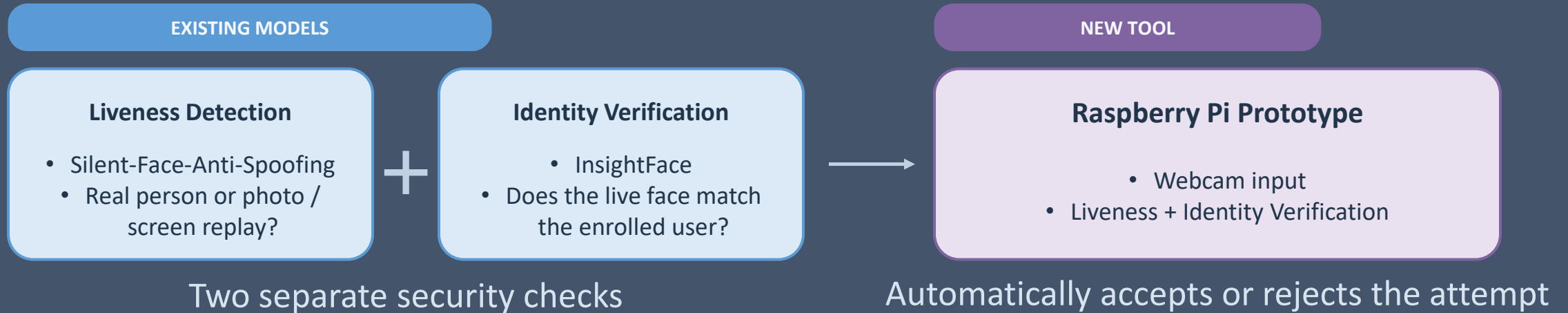
Weekly Friday check-ins • Shared GitHub • Continuous communication

Problem

- Facial recognition algorithms need to both identify if a face is valid and if it is a person's actual face as opposed to a picture of a face. Moreover, they need to do this quickly, efficiently, and accurately.
- As facial recognition becomes the standard authentication method for every-day devices, it is important that these models are both convenient and accurate
- **Central question:** are we able to quantize facial recognition models down from fp32 to int8 for battery/speed optimization while maintaining accuracy?

Solution

We will test whether INT8 quantization enables faster, lower-power facial authentication on a Raspberry Pi while keeping acceptable accuracy and security.



Compare: Latency, Memory, Energy Use, FAR/FRR/EER, etc.

Evaluation

- **Quantitative Metrics:**

- **Speed:** Measure median (p50) and 95th percentile (p95) end-to-end inference latency, covering face detection, alignment, and feature vector extraction directly on the target hardware.
- **Efficiency:** Monitor active device power draw, runtime memory consumption (Resident Set Size), and total model binary footprint on disk across precision formats.
- **Quality:** Evaluate biometric verification and identification using the core industry accuracy metrics: False Match Rate (FMR / FAR), False Non-Match Rate (FNMR / FRR), Equal Error Rate (EER), and True Accept Rate at a fixed False Accept Rate (such as TAR at FAR = 0.1%).

- **Qualitative Visualizations:**

- **ROC & DET Curves:** Plot Receiver Operating Characteristic (ROC) and Detection Error Tradeoff (DET) curves to map the operational trade-off between false matches and false rejections across different threshold settings.
- **Pareto Frontier & Disparity Plots:** Generate Pareto frontier scatterplots (Accuracy vs. Latency) to identify optimal deployment targets, and plot subgroup degradation curves across skin tone, gender, and lighting variations to evaluate Trustworthy AI fairness.

- **Baseline Comparisons:**

- **Quantization Impact:** Compare all INT8 and FP16 quantized models directly against uncompressed full-precision (FP32) baseline weights to detect where any accuracy drop-offs occur.
- **Runtime Acceleration:** Compare default framework inference against hardware-optimized runtime engines (such as ONNX Runtime with ARM NEON execution or TFLite with XNNPACK) to isolate hardware-level speedups.

Thank you

Questions?

Quantized face authentication on Raspberry Pi

